Lecture 10: Discretion vs. Commitment

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Myopic optimiser	**Dynamic optimiser**
Optimize period by period.	Optimize the sum of present and future utilities.
discretion: period-by-period optimisation without committing to future actions.	commitment: central bank can credibly announce and follow a future path. Backward induction.

Recap: Optimal Monetary Policy is strict inflation targeting.

$$\hat{\pi}_t = 0 \Rightarrow \hat{y}_t = \hat{y}_t^n = \hat{y}_t^e.$$

However, possible and optimal but not feasible since central bank can not fully stabilize inflation.

Facing short-run tradeoff due to supply shock.

NK with imperfection at the supply side

Under flexible price:

$y_t^n \neq y_t^e$ (natural output $\neq$ efficient output) $\Rightarrow$ short-run tradeoff between quantity and prices.

Move oppositely facing supply shock.

Assume at steady state, the inefficiency associated with the flexible price is not there:

$$\bar{y} = \bar{y}^n = \bar{y}^e,$$

while exists short-run deviations:

$$\hat{y}_t \neq \hat{y}_t^n \neq \hat{y}_t^e.$$

Welfare losses are:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2\right).$$

Define welfare relevant output gap as

$$\hat{x}_t = \hat{y}_t - \hat{y}_t^e = \log y_t - \log \bar{y} - \left(\log y_t^e - \log \bar{y}^{\,e}\right) = \log y_t - \log y_t^e,$$
$$\hat{\pi}_t = \log P_t - \log P_{t-1}, \qquad \alpha_x = \frac{\kappa}{\varepsilon}.$$

$$\left.\begin{aligned} &\tilde{y}_t = \hat{y}_t - \hat{y}_t^n = \hat{x}_t + (\hat{y}_t^e - \hat{y}_t^n), \\ &\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \tilde{y}_t \end{aligned}\right\} \Rightarrow \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t,$$

where

$$v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

- $E_t\{\hat{\pi}_{t+1}\}$: taken as given when choosing current-period policy.
- u_t: exogenous cost-push shock.

A nonzero u_t makes it impossible to attain simultaneously zero inflation and an efficient level of output. For example, u_t may come from exogenous changes in desired wage markups, price markups, labor income taxes, minimum wages, etc.

Assume u_t follows:

$$u_t = \rho_u u_{t-1} + \varepsilon_t^u, \qquad \rho_u \in [0, 1).$$

$$\text{NKPC:} \quad \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t,$$

$$\text{DIS:} \qquad \hat{x}_t = -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\}.$$

$\hat{r}_t^e$ is the efficient real interest rate that supports the efficient allocation.

Optimal Policy under discretion (period-by-period optimization)

$$\min_{\hat{x}_t, \hat{\pi}_t} \quad \hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 \qquad \text{s.t.} \quad \hat{\pi}_t = \kappa \hat{x}_t + v_t,$$

where

$$v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

$E_t\{\hat{\pi}_{t+1}\}$: independent of current policy; $\quad u_t$: exogenous.

$$\mathcal{L} = \hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 - \lambda(\hat{\pi}_t - \kappa \hat{x}_t - v_t).$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow 2\hat{\pi}_t = \lambda,$$
$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow 2\alpha_x \hat{x}_t = -\lambda\kappa \Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\hat{\pi}_t, \qquad \forall t \quad \text{targeting rule.}$$

To dampen the rise in inflation due to cost-push shock,

$$\hat{x}_t < 0 \Rightarrow \hat{y}_t < \hat{y}_t^e.$$

$\Rightarrow$ drive output below efficiency level; negative output gap $\Rightarrow$ policy trade-off.

Plug into NKPC:

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\hat{x}_t + u_t = \beta E_t\{\hat{\pi}_{t+1}\} - \frac{\kappa^2}{\alpha_x}\hat{\pi}_t + u_t,$$
$$\Rightarrow \quad \hat{\pi}_t = \frac{\alpha_x\beta}{\alpha_x + \kappa^2}E_t\{\hat{\pi}_{t+1}\} + \frac{\alpha_x}{\alpha_x + \kappa^2}u_t.$$

Iterate forward:

$$\begin{cases} \hat{\pi}_t = \alpha_x\psi u_t, \\ \hat{x}_t = -\kappa\psi u_t, \end{cases} \qquad \psi = \frac{1}{\kappa^2 + \alpha_x(1-\beta\rho_u)}.$$

Central bank lets output gap and inflation deviate from targets in proportion to the current value of the cost-push shock:

$$\hat{\pi}_t = \alpha_x\psi u_t, \qquad \hat{x}_t = -\kappa\psi u_t.$$

$$\psi_{0.5} = \frac{1}{\kappa^2 + \alpha_x\left(1 - \frac{\beta}{2}\right)} > \psi_0 = \frac{1}{\kappa^2 + \alpha_x}.$$

Optimal to partly accommodate inflation pressure from the cost-push shock by letting inflation rise.

Without any policy response, $\kappa = 0$:

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

Guess $\hat{\pi}_t = Au_t$ and $E_t u_{t+1} = \rho_u u_t$:

$$Au_t = \beta A\rho_u u_t + u_t \Rightarrow A = \frac{1}{1-\beta\rho_u}.$$

Without output gap response:

$$\hat{\pi}_t = \frac{1}{1-\beta\rho_u}u_t > \frac{\alpha_x}{\kappa^2 + \alpha_x(1-\beta\rho_u)}u_t.$$

$$\hat{x}_t = -\kappa\psi u_t \Rightarrow \kappa\hat{x}_t = -\kappa^2\psi u_t < 0 \quad \text{when } u_t > 0,$$
$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1}\boxed{-\kappa^2\psi u_t} + u_t.$$

u_t pushes inflation up — direct effect.

Induced recessionary output gap pushes inflation down.

Using

$$E_t\{\hat{\pi}_{t+1}\} = \alpha_x\psi\rho_u u_t,$$

$$\hat{\pi}_t = \left[\beta\alpha_x\psi\rho_u - \kappa^2\psi + 1\right] u_t = \alpha_x\psi u_t.$$

$$\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1} \Rightarrow \hat{p}_t = \hat{p}_{t-1} + \sum_{j=0}^{t} \hat{\pi}_j.$$

Since $\hat{\pi}_j = \alpha_x\psi u_j$, and $u_j \to 0$ as $j \to \infty, \Rightarrow \hat{\pi}_j \to 0$ as $j \to \infty$.

$$\Rightarrow \lim_{t\to\infty} \hat{p}_t = \hat{p}_{-1} + \sum_{j=0}^{\infty} \hat{\pi}_j > \hat{p}_{-1}, \qquad \underline{\text{level price stay permanently higher.}}$$

Optimal discretionary policy implementation

$$\hat{\pi}_t = \alpha_x\psi u_t, \qquad \hat{x}_t = -\kappa\psi u_t. \qquad \text{(Target rule)}$$

$$\hat{x}_t = -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\}.$$

Plug in:

$$-\kappa\psi u_t = -\frac{1}{\sigma}\left(\hat{i}_t - \alpha_x\psi\rho_u u_t - \hat{r}_t^e\right) - \kappa\psi\rho_u u_t,$$

$$\Rightarrow \quad \hat{i}_t = \underbrace{\hat{r}_t^e}_{\text{efficiency rate}} + \underbrace{\psi_i u_t}_{\text{response from cost-push shock}}, \qquad \psi_i = \psi\left[\kappa\sigma(1-\rho_u) + \alpha_x\rho_u\right].$$

$\kappa\sigma(1-\rho_u)$: need to affect current output gap through IS curve. $\alpha_x\rho_u$: forward-looking part.

$u_t \uparrow$ cost-push shock $\Rightarrow$ inflation $\hat{\pi}_t \uparrow \Rightarrow$ cool demand through $\hat{i}_t \uparrow > \hat{r}_t^e \Rightarrow \hat{x}_t \downarrow \Rightarrow \hat{\pi}_t \downarrow$.

But output losses are costly $\Rightarrow$ only fights $\hat{\pi}_t \uparrow$ partially.

DIS

$$\hat{x}_t = -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\}.$$

Calculation:

$$C_t^{-\sigma} = \beta(1+i_t)E_t\left[\frac{C_{t+1}^{-\sigma}}{1+\pi_{t+1}}\right].$$

At steady state, $\beta(1+\bar{i}) = 1$ and $\bar{\pi} = 0$:

$$1 = \beta(1+\bar{r}), \qquad 1+\bar{r} = \frac{1}{\beta}, \qquad \rho = -\log\beta \qquad \rho: \text{ steady-state real rate.}$$

Log-linearization:

$$\hat{c}_t = E_t\{\hat{c}_{t+1}\} - \frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\}\right],$$

$$\hat{y}_t = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\}\right] + E_t\{\hat{y}_{t+1}\}.$$

$$\hat{x}_t + \hat{y}_t^e = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\}\right] + E_t\{\hat{x}_{t+1}\} + E_t\{\hat{y}_{t+1}^e\},$$

$$\hat{x}_t = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \sigma E_t\{\Delta\hat{y}_{t+1}^e\}\right] + E_t\{\hat{x}_{t+1}\},$$

$$\hat{x}_t = -\frac{1}{\sigma}\left[\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right] + E_t\{\hat{x}_{t+1}\} \qquad \text{where } \hat{r}_t^e = \sigma E_t\{\Delta\hat{y}_{t+1}^e\}.$$

Consider an inflation targeting rule:

$$\hat{i}_t = \hat{r}_t^e + \phi_\pi\hat{\pi}_t, \qquad \phi_\pi = (1-\rho_u)\frac{\kappa\sigma}{\alpha_x} + \rho_u.$$

$$\hat{i}_t = \hat{r}_t^e + \psi_i u_t, \qquad \hat{\pi}_t = \alpha_x\psi u_t \Rightarrow u_t = \frac{\hat{\pi}_t}{\alpha_x\psi}$$
$$\Rightarrow \hat{i}_t = \hat{r}_t^e + \frac{\psi_i}{\alpha_x\psi}\hat{\pi}_t.$$

$$\frac{\psi_i}{\alpha_x\psi} = \frac{\kappa\sigma(1-\rho_u) + \alpha_x\rho_u}{\alpha_x} = (1-\rho_u)\frac{\kappa\sigma}{\alpha_x} + \rho_u = \phi_\pi.$$

Since a determinate equilibrium exists iff $\phi_\pi > 1$:

$$\phi_\pi > 1 \Rightarrow \frac{\kappa\sigma}{\alpha_x} > 1 \quad \text{may not be satisfied.}$$

A rule that decentralizes the desired discretionary equilibrium:

$$\begin{aligned}\hat{i}_t &= \hat{r}_t^e + \psi_i u_t + \phi_\pi(\hat{\pi}_t - \alpha_x\psi u_t)\\ &= \hat{r}_t^e + \Theta_i u_t + \phi_\pi\hat{\pi}_t, \qquad \text{(Instrument rule)}\end{aligned}$$

where

$$\Theta_i = \psi \left[\kappa\sigma(1-\rho_u) - \alpha_x(\phi_\pi - \rho_u) \right].$$

Now we have a unique equilibrium, and it is the desired discretionary equilibrium.

1. Under desired equilibrium:

$$\hat{\pi}_t = \alpha_x \psi u_t, \qquad \hat{i}_t = \hat{r}_t^e + \psi_i u_t.$$

We do not change the intended discretionary allocation.

2. Central bank reacts more aggressively:

$$\hat{i}_t \uparrow \quad \text{when} \quad \hat{\pi}_t > \alpha_x \psi u_t.$$

If $\phi_\pi > 1$, nominal rate rises more than one-for-one with inflation deviations; real rate rises, demand falls, inflation falls, eliminating self-fulfilling inflation paths.

Target rule: tell central bank the relationship between target variables to achieve.

E.g. $\hat{x}_t = -\dfrac{\kappa}{\alpha_x}\hat{\pi}_t$, This tells tradeoff, but still infeasible, since y_t^e is unobservable.

Instrument rule: tell central bank how to set its instrument.

E.g. $\hat{i}_t = \hat{r}_t^e + \Theta_i u_t + \phi_\pi \hat{\pi}_t$, $\hat{r}_t^e$ and u_t are hard to observe and measure.

Optimal Policy under commitment

Central bank able to commit with full credibility to a policy plan.

$$\min_{\{\hat{x}_t, \hat{\pi}_t\}_{t=0}^{\infty}} \frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t \left(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2 \right)$$

$$\text{s.t.} \quad \hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t, \qquad u_t = \rho_u u_{t-1} + \varepsilon_t^u.$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) + \gamma_t(\hat{\pi}_t - \kappa \hat{x}_t - \beta E_t\{\hat{\pi}_{t+1}\} - u_t) \right].$$

F.O.C.:

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow \hat{\pi}_t + \gamma_t - \gamma_{t-1} = 0, \qquad \gamma_{-1} = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow \alpha_x \hat{x}_t - \kappa \gamma_t = 0 \Rightarrow \gamma_t = \frac{\alpha_x}{\kappa}\hat{x}_t, \qquad \forall t.$$

Constraint in period -1 not effective for optimal plan in period 0.

$$\Rightarrow \begin{cases} \hat{x}_0 = -\dfrac{\kappa}{\alpha_x}\hat{\pi}_0, \\ \hat{x}_t = \hat{x}_{t-1} - \dfrac{\kappa}{\alpha_x}\hat{\pi}_t, \qquad t \geq 1. \end{cases} \qquad (*) \Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t.$$

$$\begin{aligned} \hat{x}_1 &= \hat{x}_0 - \frac{\kappa}{\alpha_x}\hat{\pi}_1 = -\frac{\kappa}{\alpha_x}\left(\hat{\pi}_0 + \hat{\pi}_1\right), \\ \hat{x}_2 &= \hat{x}_1 - \frac{\kappa}{\alpha_x}\hat{\pi}_2 = -\frac{\kappa}{\alpha_x}\left(\hat{\pi}_0 + \hat{\pi}_1 + \hat{\pi}_2\right). \end{aligned}$$

$$\Rightarrow \hat{x}_t = -\frac{\kappa}{\alpha_x}\sum_{s=0}^{t}\hat{\pi}_s.$$

$$\because \quad \hat{\pi}_s = \log P_s - \log P_{s-1}, \qquad \sum_{s=0}^{t}\hat{\pi}_s = \sum_{s=0}^{t}(\log P_s - \log P_{s-1}) = \log P_t - \log P_{-1}.$$

$$\Rightarrow \quad \hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t. \qquad \check{p}_t = \log P_t - \log P_{-1}, \qquad \check{p}_t: \text{ price-level targeting.}$$

While under discretion, we have purely static: $\hat{x}_t = -\dfrac{\kappa}{\alpha_x}\hat{\pi}_t$.
Now today's policy depends on past policy through $\hat{x}_{t-1}$: history dependent policy making.

Then

$$\begin{aligned} \hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa\hat{x}_t + u_t, \\ \check{p}_t - \check{p}_{t-1} &= \beta E_t\{\check{p}_{t+1} - \check{p}_t\} - \frac{\kappa^2}{\alpha_x}\check{p}_t + u_t, \\ \left(1 + \beta + \frac{\kappa^2}{\alpha_x}\right)\check{p}_t &= \beta E_t\{\check{p}_{t+1}\} + \check{p}_{t-1} + u_t. \end{aligned}$$

Let

$$a = \frac{\alpha_x}{(1+\beta)\alpha_x + \kappa^2}.$$

Then

$$\check{p}_t = a\check{p}_{t-1} + a\beta E_t\{\check{p}_{t+1}\} + au_t. \qquad \forall t$$

Assume

$$\begin{aligned} \check{p}_t &= \delta\check{p}_{t-1} + \eta u_t, \qquad \text{undetermined coefficients.} \\ \check{p}_t &= \frac{a}{\delta}(\check{p}_t - \eta u_t) + a\beta\delta\check{p}_t + a\beta\eta\rho_u u_t + au_t, \\ \left(1 - \frac{a}{\delta} - a\beta\delta\right)\check{p}_t &+ \left(\frac{a\eta}{\delta} - a\beta\eta\rho_u - a\right)u_t = 0. \end{aligned}$$

$$\begin{cases} 1 - \dfrac{a}{\delta} - a\beta\delta = 0, \\ \dfrac{a}{\delta}\eta - a\beta\eta\rho_u - a = 0, \end{cases} \Rightarrow \begin{cases} \delta = \dfrac{1 - \sqrt{1 - 4\beta a^2}}{2\beta a} \in (0,1), \\ \eta = \dfrac{\delta}{1 - \delta\beta\rho_u}. \end{cases}$$

Thus

$$\check{p}_t = \delta \check{p}_{t-1} + \frac{\delta}{1-\delta\beta\rho_u} u_t,$$

where

$$\delta = \frac{1-\sqrt{1-4\beta a^2}}{2\beta a} \in (0,1).$$

$$\hat{x}_t = -\frac{\kappa}{\alpha_x}\check{p}_t \Rightarrow \hat{x}_t = \delta \hat{x}_{t-1} - \frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)} u_t. \tag{**}$$

$$\hat{x}_0 = -\frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)} u_0.$$

$$\hat{\pi}_t = \kappa \hat{x}_t + \kappa \sum_{k=1}^{\infty} \beta^{k-1} E_t\{\hat{x}_{t+k}\} + u_t.$$

In response to u_t, central bank may lower future output gap with credible promises. Given $\hat{x}_t$, current $\hat{\pi}_t$ may decline less. Since cost function is convex, damping of deviations in period of shock improves welfare.

Assume transitory shock ($\rho_u = 0$):

$$\begin{aligned}
DIS: \hat{x}_t &= -\frac{1}{\sigma}\left(\hat{i}_t - E_t\{\hat{\pi}_{t+1}\} - \hat{r}_t^e\right) + E_t\{\hat{x}_{t+1}\} \\
\Rightarrow \quad \hat{i}_t &= \hat{r}_t^e + E_t\{\hat{\pi}_{t+1}\} + \sigma\left[E_t\{\hat{x}_{t+1}\} - \hat{x}_t\right], \\
\hat{x}_t &= -\frac{\kappa}{\alpha_x}\check{p}_t, \\
E_t\{\hat{x}_{t+1}\} - \hat{x}_t &= -\frac{\kappa}{\alpha_x}\left(E_t\{\check{p}_{t+1}\} - \check{p}_t\right), \\
E_t\{\hat{\pi}_{t+1}\} &= E_t\{\check{p}_{t+1} - \check{p}_t\}, \\
\Rightarrow \quad \hat{i}_t &= \hat{r}_t^e + E_t\{\hat{\pi}_{t+1}\} - \sigma\frac{\kappa}{\alpha_x}E_t\{\hat{\pi}_{t+1}\}, \\
\Rightarrow \quad \hat{i}_t &= \hat{r}_t^e + \left(1 - \frac{\sigma\kappa}{\alpha_x}\right)E_t\{\hat{\pi}_{t+1}\}. \qquad (1)
\end{aligned}$$

Since $\check{p}_t = \delta\check{p}_{t-1} + \delta u_t$ when $\rho_u = 0$,

$$E_t\{\check{p}_{t+1}\} = \delta\check{p}_t, \qquad E_t\{\hat{\pi}_{t+1}\} = \delta\check{p}_t - \check{p}_t = -(1-\delta)\check{p}_t.$$

Thus

$$\hat{i}_t = \hat{r}_t^e - (1-\delta)\left(1-\frac{\sigma\kappa}{\alpha_x}\right)\check{p}_t$$

Iterate backward:

$$\check{p}_t = \delta u_t + \delta^2 u_{t-1} + \cdots = \sum_{k=0}^{t} \delta^{k+1} u_{t-k}.$$

$$\Rightarrow \quad \hat{i}_t = \hat{r}_t^e - (1-\delta)\left(1-\frac{\sigma\kappa}{\alpha_x}\right)\sum_{k=0}^{t} \delta^{k+1} u_{t-k}.$$

Equilibrium nominal rate consistent with optimal allocation.

To make an implementable rule that makes the allocation unique:

$$\hat{i}_t = \hat{r}_t^e - \left[\phi_p + (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right)\right]\sum_{k=0}^{t}\delta^{k+1}u_{t-k} + \phi_p\check{p}_t, \qquad \text{ensure determinacy, } \phi_p > 0.$$

$$\Rightarrow \hat{i}_t = \hat{r}_t^e - (\phi_p + A)\check{p}_t + \phi_p\check{p}_t, \qquad A = (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right).$$

Along the intended equilibrium, the rule collapses back to

$$\hat{i}_t^* = \hat{r}_t^{e^*} - (1-\delta)\left(1 - \frac{\sigma\kappa}{\alpha_x}\right)\check{p}_t^*.$$

Without a term with $\check{p}_t$, the rule does not actively push back against the deviation.

Define deviation from the target equilibrium:

$$\tilde{p}_t = \check{p}_t - \check{p}_t^*, \qquad \tilde{x}_t = \hat{x}_t - \hat{x}_t^*, \qquad \tilde{\pi}_t = \hat{\pi}_t - \hat{\pi}_t^*, \qquad \tilde{i}_t = \hat{i}_t - \hat{i}_t^*.$$

$$\left\{\begin{aligned} \text{DIS:} \quad & \tilde{x}_t = E_t\tilde{x}_{t+1} - \frac{1}{\sigma}\left(\phi_p\tilde{p}_t - E_t\tilde{\pi}_{t+1}\right), \\ \text{NKPC:} \quad & \tilde{\pi}_t = \kappa\tilde{x}_t + \beta E_t\tilde{\pi}_{t+1}, \\ \text{Price identity:} \quad & \tilde{\pi}_t = \tilde{p}_t - \tilde{p}_{t-1}, \\ \text{Guess:} \quad & \tilde{p}_t = \lambda\tilde{p}_{t-1}. \end{aligned}\right. \Longrightarrow \left\{\begin{aligned} \tilde{x}_t &= \frac{1+\phi_p-\lambda}{\sigma(\lambda-1)}\tilde{p}_t, \\ \tilde{x}_t &= \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right]\tilde{p}_t, \\ \tilde{\pi}_t &= \left(1 - \frac{1}{\lambda}\right)\tilde{p}_t, E_t\tilde{\pi}_{t+1} = (\lambda-1)\tilde{p}_t \\ E_t\tilde{x}_{t+1} &= \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right]E_t\tilde{p}_{t+1} \\ &= \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right]\lambda\tilde{p}_t = \lambda\tilde{x}_t. \end{aligned}\right.$$

Equating the two expressions for $\tilde{x}_t$:

$$\frac{1+\phi_p-\lambda}{\sigma(\lambda-1)} = \frac{1}{\kappa}\left[(1+\beta) - \frac{1}{\lambda} - \beta\lambda\right].$$

$$H(\lambda) \equiv \sigma(\lambda-1)^2(1-\beta\lambda) - \kappa\lambda(1+\phi_p-\lambda) = 0.$$

When $\phi_p = 0$:

$H(1) = 0 \to \tilde{p}_t = \tilde{p}_{t-1} \to \tilde{p}_t$ is constant and can hold at any level deviation.

When $\phi_p > 0$:

$H(1) = \sigma(1-1)^2(1-\beta) - \kappa(1+\phi_p-1) = -\kappa\phi_p < 0 \quad \to \lambda = 1$ is not a root.

Proof that there are 1 stable root and 2 unstable roots which is what we want:

(i) For $\lambda \in (0,1)$,

$$H(\lambda) = 0 \Longleftrightarrow \frac{\kappa}{\sigma} \frac{\lambda(1+\phi_p-\lambda)}{(1-\lambda)^2(1-\beta\lambda)} = 1.$$

Then

$$LHS(0) = 0, \qquad LHS(1) = +\infty \quad \Longrightarrow \quad \exists\ \lambda_1 \in (0,1).$$

(ii) For $\lambda \in (-\infty, 0]$,

$$(\lambda-1)^2 > 0, \qquad 1-\beta\lambda > 0, \qquad -\kappa\lambda(1+\phi_p-\lambda) \geq 0.$$

$$H(\lambda) > 0 \quad \Longrightarrow \quad \text{there is no negative real root.}$$

Therefore, there are either two positive real roots or conjugate complex roots.

$$\lambda_1\lambda_2\lambda_3 = \frac{1}{\beta} > 1 \qquad \lambda_1 \in (0,1) \Rightarrow \lambda_2\lambda_3 = \frac{1}{\beta\lambda_1} > 1.$$

Thus the remaining roots are unstable:

$$\begin{cases} |\lambda_2| = |\lambda_3| > 1, & \text{if they are complex conjugate roots,} \\ \lambda_2 > 1,\ \lambda_3 > 1, & \text{if they are unstable real roots.} \end{cases}$$

Economic intuition:

(i) $\phi_p > 0$:

$$\check{p}_t \uparrow \Longrightarrow \hat{i}_t \uparrow \Longrightarrow \text{reduced demand} \Longrightarrow \text{negative output gap} \xRightarrow{\text{NKPC}} \hat{\pi}_t \downarrow.$$

Hence the belief that prices will drift up is no longer self-fulfilling; monetary policy offsets the deviation.

(ii) $\phi_p = 0$:

The central bank does not react to the endogenous price-level drift,beliefs may survive as alternative equilibria.

Distorted steady state

Permanent supply-side distortion:

$$\bar{y}^n < \bar{y}^e.$$

Real imperfections generate a permanent gap between natural and efficient levels of output:

$$-\frac{U_N}{U_C} = MPN(1-\phi), \qquad \phi = 1 - \frac{1}{\mu}.$$

Output is inefficiently low.

The welfare losses:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t \right], \qquad \text{where } \Lambda = \phi \frac{\lambda}{\varepsilon} > 0.$$

Deviation of the welfare-relevant output gap:

$$\hat{x}_t = \log X_t - \log \bar{X} = (\log Y_t - \log Y_t^e) - (\log \bar{Y} - \log \bar{Y}^e) = (\log Y_t - \log Y^e) - (\log \bar{Y}^n - \log \bar{Y}^e).$$

$$\log \bar{X} = \log \bar{Y} - \log \bar{Y}^e = \log \bar{Y}^n - \log \bar{Y}^e < 0.$$

NKPC constraint:

$$\hat{\pi}_t = \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t, \qquad \text{where } u_t = \kappa(\hat{y}_t^e - \hat{y}_t^n) \text{ is cost-push shock.}$$

i. Optimal discretionary policy

$$\min_{\hat{x}_t, \hat{\pi}_t} \frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t \qquad \text{s.t.} \quad \hat{\pi}_t = \kappa \hat{x}_t + v_t, \quad v_t = \beta E_t\{\hat{\pi}_{t+1}\} + u_t.$$

F.O.C.:

$$\hat{x}_t = \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x} \hat{\pi}_t.$$

$\frac{\Lambda}{\alpha_x}$: central bank always wants a more expansionary policy.
Plug into the constraint.

$$\hat{\pi}_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2} E_t\{\hat{\pi}_{t+1}\} + \frac{\kappa \Lambda}{\alpha_x + \kappa^2} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t.$$

Guess $\hat{\pi}_t = A + Bu_t$:

$$\Rightarrow A + Bu_t = \frac{\alpha_x \beta}{\alpha_x + \kappa^2} (A + B\rho_u u_t) + \frac{\kappa \Lambda}{\alpha_x + \kappa^2} + \frac{\alpha_x}{\alpha_x + \kappa^2} u_t.$$

$$A = \frac{\kappa \Lambda}{\kappa^2 + \alpha_x(1-\beta)}, \qquad B = \alpha_x \psi, \qquad \psi = \frac{1}{\kappa^2 + \alpha_x(1 - \beta\rho_u)}.$$

Thus

$$\Rightarrow \hat{\pi}_t^* = \frac{\Lambda\kappa}{\kappa^2 + \alpha_x(1-\beta)} + \alpha_x \psi u_t.$$

$\dfrac{\Lambda\kappa}{\kappa^2 + \alpha_x(1-\beta)}$: constant > 0; inflation bias from distorted S.S.
$\alpha_x \psi u_t$: usual response.

Changed average level of inflation / output gap; shock response no change.

Inefficient low natural level of output $\Rightarrow$

(i)positive average inflation, because CB's incentive to push output above its natural S.S. level.

(ii)increases the incentive to keep inflation higher than 0 as Λ increases (inflation bias).

ii. Optimal policy under commitment

$$\mathcal{L} = E_t \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2}(\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2) - \Lambda \hat{x}_t + \gamma_t(\hat{\pi}_t - \kappa \hat{x}_t - \beta E_t\{\hat{\pi}_{t+1}\} - u_t) \right].$$

F.O.C.:

$$\frac{\partial \mathcal{L}}{\partial \hat{\pi}_t} = 0 \Rightarrow \hat{\pi}_t + \gamma_t - \gamma_{t-1} = 0, \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial \hat{x}_t} = 0 \Rightarrow \alpha_x \hat{x}_t - \kappa \gamma_t - \Lambda = 0, \tag{2}$$

$$\gamma_{-1} = 0$$

$$(1) \Rightarrow \gamma_t = \frac{\alpha_x \hat{x}_t - \Lambda}{\kappa}, \qquad \gamma_{t-1} = \frac{\alpha_x \hat{x}_{t-1} - \Lambda}{\kappa}, \quad \text{plug into } (2) \Rightarrow \hat{x}_t = \hat{x}_{t-1} - \frac{\kappa}{\alpha_x}\hat{\pi}_t.$$

Since $\hat{\pi}_t = \check{p}_t - \check{p}_{t-1}$, by iterating forward:

$$\left.\begin{aligned} \hat{x}_t &= -\frac{\kappa}{\alpha_x}\check{p}_t + \text{constant} \\ (1) \Rightarrow \hat{x}_t &= \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x}\check{p}_t \end{aligned}\right\} \Rightarrow \text{constant} = \frac{\Lambda}{\alpha_x}$$

$$\text{Plug into NKPC:} \quad \left.\begin{aligned} \hat{\pi}_t &= \beta E_t\{\hat{\pi}_{t+1}\} + \kappa \hat{x}_t + u_t, \\ \hat{\pi}_t &= \check{p}_t - \check{p}_{t-1}, \\ \hat{\pi}_{t+1} &= \check{p}_{t+1} - \check{p}_t. \end{aligned}\right\} \Rightarrow \check{p}_t = a\check{p}_{t-1} + a\beta E_t\{\check{p}_{t+1}\} + \frac{a\kappa\Lambda}{\alpha_x} + a u_t.$$

$$\text{where } a = \frac{\alpha_x}{\kappa^2 + \alpha_x(1+\beta)}.$$

Solve with undetermined coefficients:

$$\check{p}_t = A\check{p}_{t-1} + Bu_t + C.$$

$$A = \delta = \frac{1 - \sqrt{1 - 4\beta a^2}}{2\beta a} \in (0,1), \qquad B = \frac{\delta}{1 - \delta\beta\rho_u}, \qquad C = \frac{\delta\kappa\Lambda}{(1-\delta\beta)\alpha_x}.$$

Thus
$$\check{p}_t = \delta \check{p}_{t-1} + \frac{\delta}{1-\delta\beta\rho_u} u_t + \frac{\delta\kappa\Lambda}{(1-\delta\beta)\alpha_x}. \tag{3}$$
$\dfrac{\delta}{1-\delta\beta\rho_u} u_t$: shock response, not affected.
$\dfrac{\delta\kappa\Lambda}{1-\delta\beta}$: stabilization bias; shifts price-level path upward by a constant
$\Rightarrow$ price level converges to a constant and average inflation is 0.

$$\lim_{T\to\infty} P_T = P_{-1} + \frac{\delta\kappa\Lambda}{(1-\delta\beta)(1-\delta)\alpha_x}.$$

Zero average inflation, but positive upward bias above P_{-1}.

$$\hat{x}_t = \frac{\Lambda}{\alpha_x} - \frac{\kappa}{\alpha_x} \check{p}_t.$$

Plug in (3):

$$\hat{x}_t = \underbrace{\delta \hat{x}_{t-1}}_{\text{history dependence}} - \underbrace{\frac{\kappa\delta}{\alpha_x(1-\delta\beta\rho_u)} u_t}_{\text{same shock response}} + \underbrace{\frac{\Lambda}{\alpha_x}\left[1-\delta\left(1+\frac{\kappa^2}{(1-\delta\beta)\alpha_x}\right)\right]}_{\text{constant S.S. deviation}}.$$

Commitment leads to:

1. output can be raised above its natural level with welfare improvement;
2. more subdued effects on inflation;
3. commitment avoids the inflation bias under discretionary policy.